

AI Exposure or Occupational Composition?

Constructed Measures and Early-Career Employment

Lily Luo

[AUTHOR TO COMPLETE: affiliation]

[AUTHOR TO COMPLETE: email]

September 2026

Abstract. I rebuild a widely discussed young-worker employment pattern in public Current Population Survey data and ask whether detailed occupational AI exposure carries information beyond broad occupational composition. In the corrected January 2017–July 2026 panel, the post-January-2023 young-to-older employment-stock contrast between the highest and lowest Eloundou exposure quintiles is -0.1346 log points. Conditioning on two-digit occupation-family-specific post changes reduces it to -0.0315 (95-percent interval $[-0.1676, 0.1046]$); allowing each family its own monthly young-relative path gives -0.0317 . The reduction is statistically detectable in a paired comparison, but the conditional models retain only 30 percent of baseline target information and change the estimand. Food-service exclusions explain little of the reduction. Architecture comparisons are less informative than their point estimates suggest: the baseline design has an 80-percent minimum detectable effect of 0.1244 log points, and paired differences across seven alternative constructions all include zero. A public implementation of Brynjolfsson–Chandar–Chen’s top-two-versus-bottom-three exposure grouping yields a smaller -0.0728 CPS contrast. Taxonomy repair and construction choices remain consequential, but the defensible conclusion is narrower: the CPS association is principally a broad-occupation comparison, not an identified effect of detailed AI susceptibility.

JEL codes: J23, J24, O33, C81.

Keywords: artificial intelligence, occupational exposure, occupational composition, early-career employment, constructed treatments.

Acknowledgments: [AUTHOR TO COMPLETE].

1 Introduction

Recent evidence has linked occupational exposure to generative artificial intelligence with weaker employment growth among workers in their early twenties. The pattern matters because it bears on whether new technology is already changing entry-level work. It is also unusually hard to interpret. Occupational exposure is fixed before the labor outcome is observed, is highly correlated with broad occupational structure, and is available in several constructed forms that do not necessarily measure the same object. A negative exposed-versus-unexposed contrast can therefore reflect detailed AI susceptibility, broad occupational composition, or both.

This paper asks a deliberately narrower question: *does a detailed occupational AI score explain the public-CPS early-career employment pattern beyond the evolution of broad occupation families?* I aggregate Basic Monthly Current Population Survey records into occupation-by-age-by-month employment stocks from January 2017 through July 2026. The design compares ages 22–25 with ages 26–65 within occupations and asks whether that relative stock changes after January 2023 across exposure groups. The outcome is a stock, not hiring, and the design is descriptive. It does not observe employer adoption and does not turn the release of ChatGPT into an instrument.

The corrected-calendar baseline reproduces a sizeable pattern. The post-January-2023 young-relative coefficient for the highest versus lowest employment-weighted quintile of Eloundou et al.’s GPT-4 beta exposure is -0.1346 log points, with a canonical 9,999-draw 95-percent wild-score interval of $[-0.2223, -0.0468]$. The frozen 108-month chronology gives -0.1311 and its canonical interval $[-0.2171, -0.0451]$. These numbers describe the change in a young-to-older employment-stock ratio, not a percentage decline in individual employment probability.

The central result changes the interpretation. Adding two-digit-occupation-family by young by post interactions reduces the corrected coefficient to -0.0315 with interval $[-0.1676, 0.1046]$. Allowing every broad family its own young-relative monthly path gives -0.0317 . The paired baseline-to-conditioned movement is about 0.103 log points and its interval excludes zero. Yet this is not an additive decomposition: the conditioning model asks whether detailed quintile status predicts outcomes among the comparisons left after broad-family evolution is absorbed. Only four of 22 broad families contain both Q1 and Q5, and conditional target information falls to 30 percent of the baseline. The evidence says that the baseline is predominantly a comparison across broad occupation families and that the remaining within-family association is weakly measured. It does not assign a causal share to broad composition.

Nor does a simple food-service story account for the result. Excluding low-exposure food-

service occupations attenuates the coefficient by 8.8 percent, to -0.1196 . Excluding food, personal-care, and protective-service occupations makes it slightly more negative. Jointly removing the five occupations with the largest prior leave-one-out movements yields -0.1011 , while symmetric trimming and continuous down-weighting return estimates close to baseline. Influence is consequential and signed in both directions; “food-service recovery explains the pattern” is not supported.

The second result concerns precision. The realized occupation-clustered standard error is 0.0444 , compared with 0.0122 in the planning simulation. The frozen design’s normal-theory minimum detectable effect at 80-percent power is 0.1244 log points, approximately the size of the headline coefficient. Paired minimum detectable differences across alternative exposure constructions range from 0.0609 to 0.1689 . Every paired Q5–Q1 interval comparing beta with an alternative contains zero. The data therefore do not detect architecture differences at conventional levels; they also do not establish that the architectures are economically equivalent. This distinction replaces earlier language that treated smaller or insignificant alternative point estimates as evidence that the conclusion did not generalize.

The third result connects the public-data exercise directly to [Brynjolfsson, Chandar, and Chen \(2026\)](#). Implementing their published GPT-4 beta classification—the top two exposure quintiles against the bottom three—inside the same CPS model yields -0.0728 (interval $[-0.1240, -0.0216]$). This is a reproducible component of their design, not a replication of proprietary ADP outcomes, job-title mapping, firm controls, hiring, or separation. Reapplying the grouping across exposure architectures produces coefficients between approximately -0.088 and -0.008 on common support; only beta’s difference from the OECD capability measure is detected under that grouping. The contrast itself is part of the empirical object.

Measurement still matters, but in more specific ways than the earlier manuscript claimed. A literal SOC-2010 exact-code merge into post-2018 data covers only 3.33 percent of computer-and-mathematical employment; the documented taxonomy repair covers 97.7 percent. Most of the associated coefficient change comes from readmitting occupations rather than changing exposure values on fixed support. The original six implementations also span little independent variation: two weighted principal components explain 96.11 percent of their common-support variance. These facts motivate a taxonomy that separates implementation errors, same-construct measurement changes, construct changes, support changes, estimand changes, representation changes, and pure reparameterizations.

The paper contributes a public-data diagnosis of a prominent descriptive pattern and a disciplined way to audit constructed treatments. Its substantive conclusion is negative but informative: in the CPS, detailed AI quintiles do not carry a precisely estimated association after broad occupational evolution is absorbed. Its measurement conclusion is equally bounded:

alternative constructed scores can differ in coverage and classification while available outcomes remain too noisy to distinguish many downstream coefficients. The paper does not estimate a causal AI effect, infer economic equivalence from a confidence interval containing zero, or claim that broad-family conditioning identifies the non-AI counterfactual.

The remainder of the paper defines the construction layers, documents the corrected data and inferential target, presents the broad-family result, and then evaluates architecture precision, reference groups, influence, population controls, and timing. Mobility and algebraic reparameterizations are demoted to the online appendix because they do not resolve the employment-stock question.

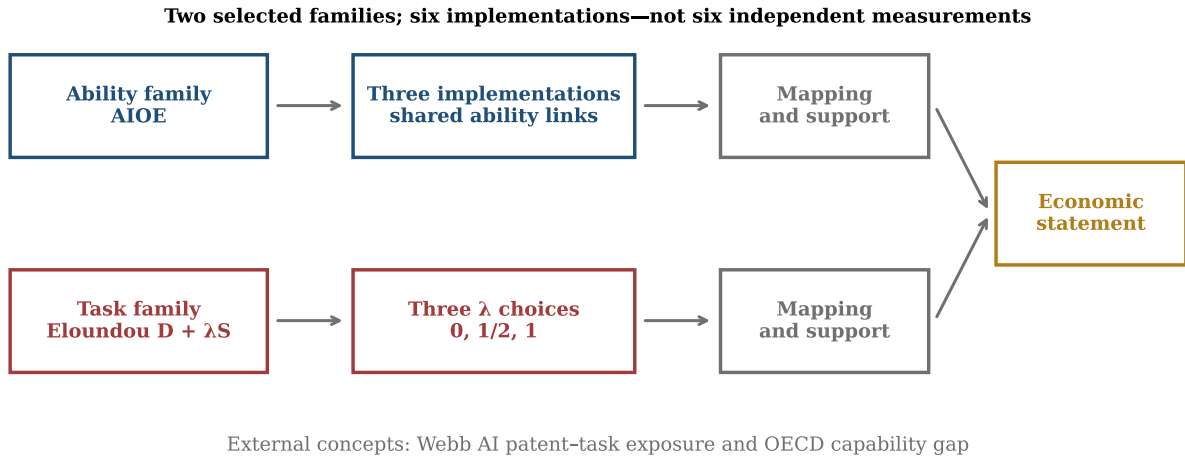


FIGURE 1. From exposure construction to an economic statement

Notes: The six initial implementations belong to two selected families. Webb AI and the OECD capability gap change the construct as well as the score. Mapping, support, representation, comparison, and conditioning determine the downstream estimand. Potential exposure is not observed adoption.

2 Constructed Treatments and Related Evidence

2.1 A layered empirical object

Write a constructed-treatment analysis as a sequence from an economic construct C , through a measurement rule M , a harmonization rule H , a retained support Ω , and a representation R , to an estimand Q :

$$C \xrightarrow{M} x^N \xrightarrow{H} x \xrightarrow{\Omega, w} x|_{\Omega} \xrightarrow{R} Z \xrightarrow{Q, P} \theta.$$

Calling every perturbation a different “measure” conceals economically important distinctions. An invalid taxonomy merge is an implementation error. Reweighting the same application–

ability links is a same-construct measurement change. Moving from LLM task acceleration to patent overlap or a broad capability gap changes the construct. Expanding mapped occupations changes the target population. Converting a level to quintiles changes the treatment representation. Replacing Q5–Q1 by Q5–Q2 changes the comparison. Rotating a fixed two-regressor column space changes only coordinates.

Table 1 gives the reporting taxonomy used throughout. A row can carry more than one label: repairing a mapping both corrects an implementation and may readmit occupations. The analysis separates the fixed-support score change from the support expansion rather than calling their sum a “crosswalk effect.”

Two simple invariance results discipline the exercise. First, for fixed positive weights and a fixed tie rule, a positive affine transformation preserves ranks and weighted-quantile membership. Second, if two regressor matrices are related by a nonsingular coordinate transformation and sample, objective, nuisance terms, and constraints remain fixed, fitted values and transformed contrasts are identical. The F/G to A/E exercise in the prior draft is therefore a basis change, not additional empirical corroboration; it is retained only in the online appendix.

The task-family construction has an equally useful structure. If D_o is the share of tasks directly accelerated by an LLM and S_o the share accelerated only with complementary software, then

$$X_{\alpha o} = D_o, \quad X_{\beta o} = D_o + \frac{1}{2}S_o, \quad X_{\gamma o} = D_o + S_o. \quad (1)$$

For any pair of occupations, rankings along $D + \lambda S$ can change only where their affine score paths cross. Weighted-bin assignments are consequently piecewise constant in λ on fixed support. There is no reason, however, for estimated categorical coefficients to be monotone in λ .

2.2 Exposure architectures

The core audit contains three AIOE implementations and three Eloundou implementations. AIOE links progress in ten AI applications to 52 O*NET abilities and then to occupational ability requirements (Felten, Raj, and Seamans 2018, 2021). Administrative, direct-ability, and source-employment-weighted versions vary reconstruction or aggregation within that system. The Eloundou family changes the technology boundary through equation (1) (Eloundou et al. 2024). These are six implementations from two families, not six independent measurements of a known latent exposure.

The treatment-side dependence is strong. On the exact 444-occupation common support,

TABLE 1. Which layer of a constructed treatment changes?

Operation	What changes	Clean comparison requires	Interpretation in this paper
Implementation re-pair	Executed code, source vintage, or mapping violates its declared rule	Identify the corrected implementation first	Exact-code matching across incompatible SOC vintages is an error, not a defensible rival estimate
Same-construct measurement	Labels, source inputs, or aggregation within a maintained construct	Fixed support, representation, estimand, and controls	Three AIOE reconstructions compare implementations of one ability/application system
Construct change	Technology boundary, primitive, horizon, or potential exposure versus use	A common downstream design	Eloundou task acceleration, Webb patent overlap, and the OECD capability gap ask different economic questions
Support or population	Occupations, dates, or weights entering the target population	Fixed scores and representation on the overlap	Taxonomy repair mostly readmits occupations; native- and common-support estimates target different populations
Estimand or representation	Outcome contrast, comparator, conditioning set, ranks, bins, or continuous level	All upstream layers fixed	Q5–Q1, BCC top-two versus bottom-three, and a continuous slope are not the same parameter
Pure reparameterization	Coordinates change but the fitted column space does not	Same sample, objective, regressors, and constraints	The F/G to A/E rotation is an identity, not new evidence

Notes: A comparison can change more than one layer. The paper reports the direct operation and induced changes separately. Positive affine transformations preserve ranks and fixed-weight quantile groups; full-rank coordinate changes preserve fitted values after the coefficient vector is transformed.

the first weighted principal component explains 80.04 percent of standardized-score variance and the first two explain 96.11 percent; three explain 99.58 percent. The intersection–union sign calculation for the six coefficients is therefore retained only as a predeclared statement about a dependent set. It is not presented as six independent confirmations, and the failed simultaneous-band criterion remains in the design record.

Two externally admitted constructions use different primitives. Webb maps patent text to occupational tasks, while the OECD measure compares nine capability domains with occupational requirements (Webb 2020; Organisation for Economic Co-operation and Development 2026). They are different economic constructs, not reliability draws for the same LLM variable. Frey–Osborne remains an automation comparator rather than being relabeled as AI. A repository-bounded census records six core implementations from two families, two admitted external implementations from two additional families, one scope failure, and several published candidates whose score artifacts or mappings are not yet locally verified. This is not claimed to be a timeless census of the literature.

2.3 Relation to labor-market evidence

Brynjolfsson, Chandar, and Chen (2026) use proprietary ADP payroll records and emphasize employment declines among ages 22–25 in highly exposed jobs, operating largely through hiring. Their public rule places GPT-4 beta occupations in employment-weighted quintiles and contrasts the top two with the bottom three. Section 5 implements that public classification in CPS. It cannot recover their employer panel, title-to-occupation mapping, firm-time controls, hiring, or separations.

Other studies answer different questions. Humlum and Vestergaard (2025) link reported adoption to Danish administrative outcomes; workplace deployments observe actual tool access rather than occupational potential (Brynjolfsson, Li, and Raymond 2025). These designs distinguish exposure from use but do not validate one universal occupation score. The broader task literature likewise emphasizes substitution, complementarity, and occupational adjustment rather than a one-dimensional technology dose (Autor, Levy, and Murnane 2003; Autor and Dorn 2013; Acemoglu and Restrepo 2019).

This paper is closest in spirit to sensitivity analysis that asks which moments or trends support an empirical result (Andrews, Gentzkow, and Shapiro 2017; Rambachan and Roth 2023) and to work on generated covariates (Battaglia et al. 2024; Christensen and Hansen 2026; Duan and Pelger 2026; Ludwig, Mullainathan, and Rambachan 2026). Conventional occupation-clustered inference here conditions on labels, mapping, and support. The online appendix specifies what a validation sample would need to contain before generated-covariate

corrections could address occupational AI exposure itself.

3 Data, Calendar, and Occupational Harmonization

3.1 CPS stocks and the corrected calendar

The public-data application uses IPUMS Basic Monthly CPS records from January 2017 through July 2026. Employed respondents with positive final weights are aggregated into occupation-by-age-group-by-month stocks. Ages 22–25 form the young group and ages 26–65 the comparison group. Age is not labor-market experience, and the outcome can change through employment entry, exit, occupational switching, or the older denominator.

The original wide extract accidentally selected the March ASEC sample ('03s') rather than the Basic Monthly sample ('03b') in 2017–2021. ASEC records carry zero Basic Monthly final weights for this purpose. A separately authenticated extract restores those five March samples. October 2025 remains absent because no CPS was collected during the federal shutdown. The combined files contain 9,843,021 records and 114 observed survey months. Omitting December 2022 as a transition leaves 113 months in the substantive static model.

The corrected calendar is now the substantive baseline. Restoring March changes the coefficient from -0.1311 to -0.1346 , with a 9,999-draw interval $[-0.2223, -0.0468]$. The frozen result remains visible as the confirmatory chronology record but no longer defines the default exposition.

3.2 Taxonomy bridge and universe reconciliation

CPS occupations use Census-2010 codes through 2019 and Census-2018 codes thereafter, while source exposures use SOC vintages. Pre-2020 records are expanded along official Census-2010-to-2018 routes and aggregated to the common target taxonomy. In a one-to-many route, official conversion proportions are applied equally to young and older source records. This preserves the source occupation's age ratio across target components by construction; official files do not supply age-specific probabilities.

The full route-expanded preperiod universe contains 539 target occupations with positive young and older stock. Two counts that appeared to be a five-occupation discrepancy are instead nonnested, architecture-specific supports. The stored 490-occupation universe requires full beta coverage. The 495-occupation continuous-mapping universe requires repaired AIOE and Webb coverage. They overlap in 466 occupations: 24 appear only in beta support and 29 only in AIOE-plus-Webb support. Adding Webb to beta leaves the 468-occupation primary

model. The online appendix names every member and reason.

Within primary support, split sources contribute 14.93 percent of young and 17.75 percent of older weighted stock during 2017–2019. Ninety-nine target occupations can receive a split route and carry 23.38 percent of fitted target information; proportional allocation of information by split-stock share assigns 4.44 percent to split stock. The former is a conservative touched-target diagnostic and the latter is not an identified decomposition.

The negative association does not depend on split routes. Restricting the corrected model to 369 targets with no structurally possible split inbound route gives -0.1674 under fixed labels and -0.1471 after recomputing the comparison on that support. Aggregating to a stable Census-2010 taxonomy gives -0.1522 . Each changes the population and should not be called a correction of the primary estimate.

Age-specific route allocation remains uncertain. Under unrestricted official-route accounting, the 2017–2019 young/older stock ratio can range from .1031 to .1480 in Q1 and from .0790 to .1175 in Q5. These are sharp accounting ranges for stocks, not bounds on the nonlinear regression coefficient. A predeclared tilt that changes young-versus-older allocation odds toward the highest-beta eligible target by factors from .5 to 2 moves the coefficient from -0.1292 to -0.1391 . Only 6.88 percent of early stock is eligible for that scenario, so it is not a global robustness claim.

3.3 A detectable workflow failure

Literal exact-code matching of SOC-2010 AIOE into a post-2018 outcome taxonomy covers just 3.33 percent of computer-and-mathematical employment. The documented bridge reaches 97.7 percent. A machine-readable compatibility gate now rejects a merge when declared source and target vintages are not identical and no versioned route is supplied.

The corresponding continuous sequence separates layers: original score/original 410-occupation support gives -0.01885 ; repaired score/same support gives -0.01920 ; repaired score/expanded 495-occupation support gives -0.03156 ; excluding computer/math gives -0.02940 . The invalid exact-code result is not an economically defensible alternative. The small first movement concerns score correction; the larger second movement is population re-admission.

3.4 Treatment formation

The primary Eloundou beta score and Webb software control are standardized using preperiod employment weights. Beta is divided into employment-weighted quintiles with ties retained.

On the 468-occupation primary support, Q1 contains 133 occupations and 20.02 percent of employment; Q5 contains 91 and 19.53 percent. All support and sensitivity rows state whether cutoffs and scales are frozen or recomputed. Respondent-equivalent counts after route expansion are constructed records, not unique people.

TABLE 2. Reconciliation of occupation universes

Universe	Occupations	Months used to form support	Availability rule
Raw route-expanded balanced candidates	539	66	Positive young and older preperiod stock
Frozen beta-complete support	490	66	Raw balance plus finite beta
Repaired AIOE plus Webb support	495	66	Raw balance plus finite repaired AIOE and Webb
Intersection of preceding two supports	466	66	Satisfies both architecture-specific rules
Primary beta plus Webb model	468	108/113	Finite beta and Webb in frozen/corrected static panel
Six core measures plus Webb	444	108	Literal complete-case intersection

Notes: The 490- and 495-occupation supports are not nested: 24 occupations occur only in beta-complete support and 29 only in repaired-AIOE-plus-Webb support. The first four rows define preperiod availability; the final two are fitted-model samples after the full calendar and model-specific requirements. Frozen and corrected calendars contain 108 and 113 static-model months, respectively.

4 Estimand and Inference

4.1 Employment-stock contrast

Let N_{oat} be the CPS-weighted employment stock in occupation o , age group a , and month t . Let Y_a identify ages 22–25, P_t identify January 2023 onward, Q_{oq} denote exposure quintile q , and W_o denote standardized Webb software exposure. The maintained mean is

$$E[N_{oat}|X] = \exp \left[\gamma_{oa} + \delta_{ot} + \lambda_{at} + \sum_{q=2}^5 \beta_q Q_{oq} Y_a P_t + \theta W_o Y_a P_t \right]. \quad (2)$$

Occupation-by-age, occupation-by-month, and age-by-month fixed effects absorb time-invariant age composition within occupation, common occupation shocks, and aggregate age-specific shocks. The target β_5 is the change in the young-to-older employment-stock ratio for Q5 relative to Q1 after January 2023, conditional on Q2–Q4 and Webb software exposure. It is not an individual employment probability, a hiring rate, or a continuous exposure slope.

With two age groups, conditioning independent Poisson pseudo-likelihood contributions on $T_{ot} = N_{oyt} + N_{oot}$ gives

$$\ell_{ot} = N_{oyt} \log p_{ot} + N_{oot} \log(1 - p_{ot}), \quad \text{logit}(p_{ot}) = \log(\mu_{oyt}/\mu_{oot}). \quad (3)$$

The occupation-month effect cancels. CPS final weights construct the two continuous cell stocks and are not applied a second time to the objective. The grouped-binomial expression is an estimating equation for weighted stocks, not a literal model of independent survey-weighted trials.

The central conditioning exercise augments equation (2) with SOC2-family-specific $Y_a P_t$ terms. A saturated version instead absorbs SOC2-family-specific young-relative effects in every month. These models ask whether detailed exposure groups predict outcomes after broad-family evolution is conditioned out. They do not decompose the baseline coefficient into causal AI and non-AI components, because the conditioning set and therefore the estimand change.

4.2 Maintained source of randomness

The principal confidence interval treats occupation-level economic score contributions as independent across 468 occupation clusters and permits arbitrary serial dependence within occupation. It conditions on the observed CPS-weighted cells, constructed exposure labels, taxonomy mapping, support, quintile cuts, and fitted specification. Thus it is model-based uncertainty for a stochastic occupational-shock parameter. It is not a complete design-based CPS variance estimator and does not cover exposure-label generation, taxonomy-allocation uncertainty, or every dependence induced by respondents moving across occupations.

For transparency, let x_{ot} be the target and nuisance regressors after the grouped-binomial fit, p_{ot} the fitted young share, and $h_{ot} = T_{ot} p_{ot} (1 - p_{ot})$. The implementation absorbs the fixed effects from x_{ot} under weights h_{ot} , producing $\tilde{x}_{ot}$. It forms the information matrix

$$H = \sum_{o,t} h_{ot} \tilde{x}_{ot} \tilde{x}_{ot}',$$

the occupation score $s_o = \sum_t \tilde{x}_{ot}(N_{oyt} - T_{ot}p_{ot})$, and the one-step influence $H^{-1}s_o$, with the finite-cluster factor $\sqrt{G/(G-1)}$. Each wild-score draw multiplies the entire occupation influence vector by one Rademacher sign. The analytic occupation-cluster standard error studentizes the centered draw; the 95-percent critical value is the empirical .95 quantile of the absolute studentized shifts. Paired comparisons reuse the same signs for both fits, thereby preserving estimated covariance. The procedure does not create pseudo-outcomes or refit each draw.

All central revised intervals use 9,999 multipliers. The sole canonical interval for the frozen -0.1311 estimate is $[-0.2171, -0.0451]$; previously displayed variants came from different finite sets of 999 multipliers, except a minimum-count row that changed the sample. Pair-specific member intervals remain labeled as such and never replace the canonical interval.

4.3 Precision, sparse cells, and rotation sensitivity

The realized primary SE is 0.0444, versus 0.0122 in the planning simulation. Its normal-theory 80-percent minimum detectable effect is therefore 0.1244 log points, or 13.25 percent on the relative-stock-ratio scale. The paper uses MDEs to characterize what the design can resolve; it does not substitute an arbitrary smallest effect of interest or interpret a confidence interval containing zero as equivalence.

Young cells are sparse: the 10th, 50th, and 90th percentiles of respondent-equivalent counts are 0, 2, and 16; 26.3 percent are zero and 69.9 percent are below five. Older-cell percentiles are 4, 25, and 199. One-sided cells contribute valid finite terms to equation (3); only 965 occupation-months with both stocks zero are omitted. Conditioning on at least five observed young respondents attenuates the coefficient to about -0.109 , but this selects on realized outcomes, retains fewer than half of occupation-months, and changes the estimand.

Quarterly aggregation gives -0.1310 under the frozen chronology and -0.1345 under the repaired chronology, essentially matching monthly estimates and SEs. To probe CPS rotation and common monthly shocks, a post-outcome score-covariance sensitivity adds contemporaneous and lagged covariance in aggregate month scores while subtracting the occupation-month intersection. At a 12-month Newey–West lag the SE rises to 0.0491 and the normal interval is $[-0.2273, -0.0349]$. Occupation clustering already permits arbitrary within-occupation serial dependence; this extra calculation addresses declared cross-occupation time covariance but remains model-based rather than a CPS replicate-weight variance.

Finally, a historical simulation that preserves each donor month’s entire contemporaneous cross-occupation residual vector under a sharp global-sign design closes none of the prospective-to-realized precision gap. It rules out that specific implementation as the explanation; it does

not validate the planning calculation or identify which unmodeled dependence generated the gap.

5 Where the Employment Association Comes From

5.1 Baseline and broad occupation families

The substantive baseline uses the repaired 113-month calendar and frozen beta treatment definition. The Q5–Q1 coefficient is -0.1346 with occupation-clustered SE 0.0450 and canonical 95-percent interval $[-0.2223, -0.0468]$. The frozen design record remains -0.1311 with the canonical interval $[-0.2171, -0.0451]$. Calendar repair therefore does not explain the association.

The direct regression analogue of the earlier within-SOC2 permutation does. Adding broad-family-specific young-relative post terms reduces the coefficient to -0.0315 ; allowing each family its own young-relative monthly path gives -0.0317 . Both intervals include zero. On common draws, the SOC2-by-post estimate minus baseline is $+0.1031$, with paired interval $[0.0035, 0.2026]$; the monthly version gives $+0.1028$, with interval $[0.0035, 0.2021]$. Thus the movement is detected even though the residual coefficient is not.

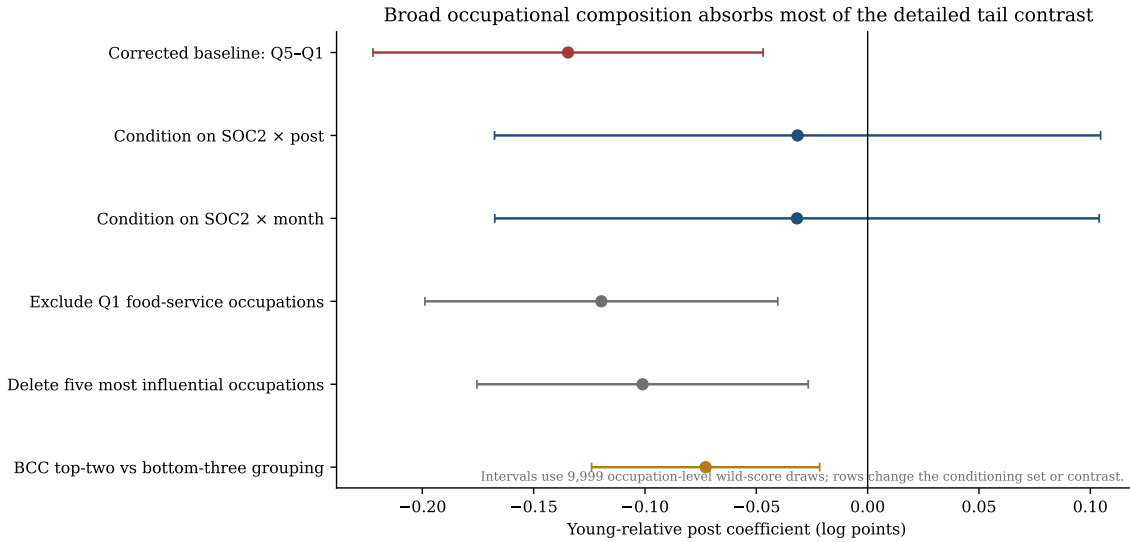
This is strong evidence about the comparison and weak evidence about a structural mechanism. In point-estimate terms, conditioning absorbs about 76 percent of the baseline magnitude. But the target changes from a cross-economy Q5–Q1 contrast to a coefficient assembled from comparisons remaining within broad families after their differential evolution is removed. Only four of 22 SOC2 families contain both Q1 and Q5, and conditional target information falls from 26.4 million to 8.0 million, about 30 percent of baseline. The conditional coefficient is connected through intermediate quintiles in most families. It should not be described as “the remaining AI effect,” and the difference should not be described as the causal contribution of broad composition.

The prior within-SOC2 permutation is consistent with this result but is no longer asked to carry it. Randomly reassigning detailed beta scores within broad families produces a mean coefficient of -0.1097 against the frozen -0.1311 . Its exchangeability assumption is not established, so the permutation percentile is descriptive. The conditioned regressions state the relevant comparison directly and show the support and information cost.

TABLE 3. Broad occupational composition and the corrected employment contrast

Specification	Coef.	SE	95% interval	Info. share	Occ.
Corrected 113-month Q5–Q1 baseline	-.1346	.0450	[-.2223, -.0468]	1.000	468
Add SOC2 $\times$ young $\times$ post	-.0315	.0706	[-.1676, .1046]	.301	468
Add SOC2 $\times$ young $\times$ month	-.0317	.0703	[-.1674, .1040]	.300	468
Exclude Q1 food-service occupations	-.1196	.0406	[-.1988, -.0404]	—	463
Exclude all SOC35/37/39 in-person services	-.1370	.0435	[-.2217, -.0523]	—	435
Delete five most influential occupations	-.1011	.0385	[-.1755, -.0267]	—	463
Stable tails under all six implementations	-.2120	.1030	[-.4131, -.0109]	—	64

Notes: All rows are post-outcome exploratory and use 9,999 occupation-level wild-score draws. The baseline row uses its canonical draw set; conditioned and diagnostic rows use their declared common-draw sets. The first three retain the same 468 occupations and treatment groups. “Info. share” is conditional target information relative to the corrected baseline. Adding SOC2 interactions changes the conditioning estimand; the coefficient reduction is not an additive or causal decomposition. Only four of 22 broad families contain both Q1 and Q5. Exclusions retain frozen groups. The stable-tail row compares 46 always-Q1 and 18 always-Q5 occupations and has an effective information count of 15.0.

**FIGURE 2.** Broad occupational composition and alternative comparisons

Notes: All rows are post-outcome exploratory and use 9,999 occupation wild-score draws. SOC2 rows change the conditioning estimand. Food and influence rows retain frozen groups but change support. The BCC row changes the exposure-group contrast. Intervals are therefore informative within rows but the coefficients are not estimates of one invariant parameter.

5.2 Profile, reference tail, and stable classifications

The fitted post coefficients relative to Q1 are 0, -0.0855 , -0.0478 , -0.0970 , and -0.1311 . A three-restriction test does not reject a common coefficient for Q2–Q5 ($p = .119$), but cannot establish their equality. The visible Q3–Q2 increase also does not yield a rejection of monotone decline under a least-favorable max- t test ($p = .393$). The data therefore neither establish nor reject monotonicity.

Changing the requested contrast clarifies Q1’s role. Q5–Q2 is -0.0456 with interval $[-0.1248, 0.0335]$; Q5–Q4 is -0.0340 with interval $[-0.1110, 0.0429]$; Q5–Q3 is -0.0833 with interval $[-0.1581, -0.0084]$. These are different economic contrasts, not alternative omitted-category parameterizations. Figure 3 shows why: both the low-exposure trajectory and the older-worker denominator matter. The fitted contrast is not simply a collapse of young employment in Q5.

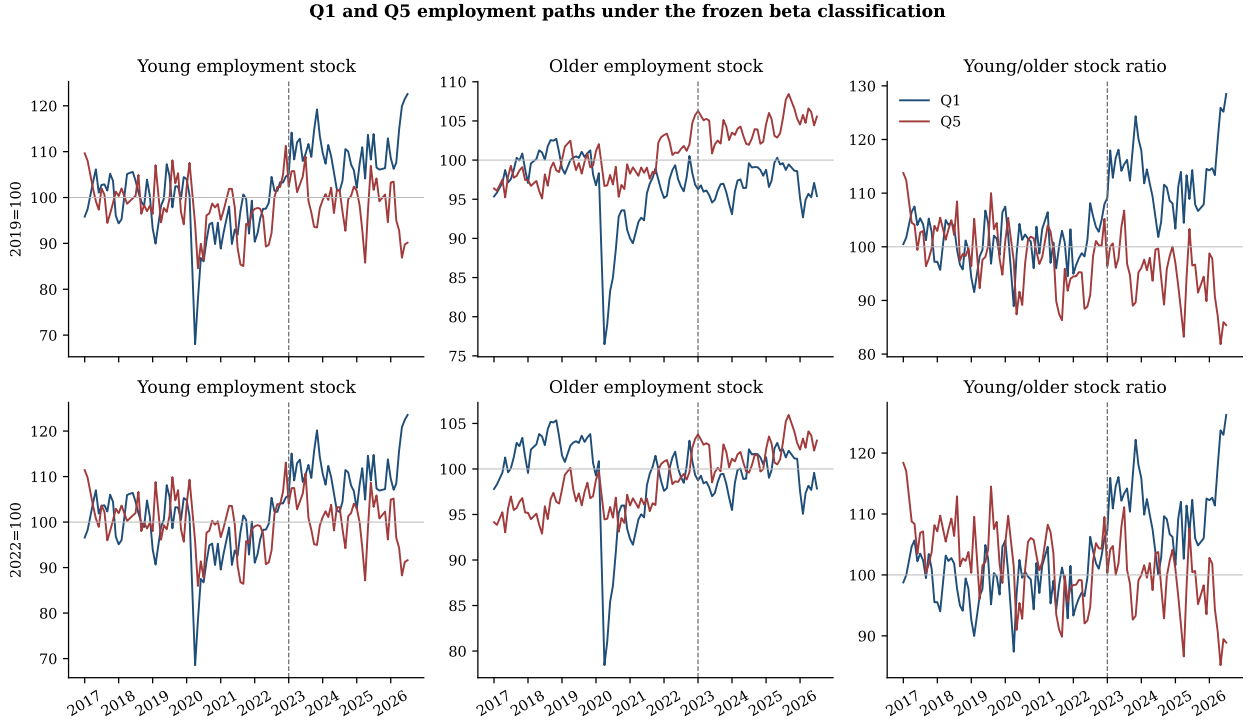


FIGURE 3. Q1 and Q5 employment stocks and ratios

Notes: Monthly CPS-weighted stocks under frozen beta membership. The upper row indexes to observed 2019 means and the lower row to observed 2022 means. The dashed line marks January 2023. October 2025 is absent because no CPS was collected. The paths are descriptive and do not additively decompose the nonlinear coefficient.

Only 46 occupations are always Q1 and 18 always Q5 across the six original implementations. These stable tails account for 9.74 percent of common-support employment. Their contrast is -0.2120 with interval $[-0.4131, -0.0109]$, but has only 15.0 effective information occupations

and a 46.7-percent top-five information share. The negative estimate survives in this narrow population; it does not show that unstable classification is innocuous in the broader one.

5.3 Influence and the food-service hypothesis

Influence is economically consequential but not one-directional. Jointly deleting the five occupations with the largest absolute prior leave-one-out movements reduces the estimate to -0.1011 . Deleting the top ten or twenty instead makes it more negative, -0.1522 and -0.1555 , because later occupations have opposing signed influence. A symmetric 2.5-percent trim of each signed-influence tail gives -0.1281 , and Huber-style continuous down-weighting gives -0.1314 . These are outcome-adaptive diagnostics rather than preferred specifications.

Cleaner occupation-side exclusions do not support the strong claim that food-service recovery explains the headline. Dropping the five Q1 food-service occupations yields -0.1196 , an 8.8-percent reduction in absolute magnitude. Dropping all food-service occupations gives -0.1201 . Dropping all food, building-and-grounds, personal-care, and protective-service occupations yields -0.1370 ; dropping only their Q1 members gives -0.1387 . These exercises cannot isolate reopening, immigration, minimum wages, or labor supply, but they reject the simpler assertion that removing in-person service occupations makes the pattern disappear.

6 What Exposure Construction Changes—and What the Data Resolve

6.1 Paired architecture precision

Alternative exposure constructions change occupation ranks, tail membership, and point estimates. The key inferential question is whether the downstream coefficient differences are resolved once both estimates use common support and common occupation multipliers. Table 4 answers no for the original Q5–Q1 comparison. Every beta-minus-alternative paired interval contains zero. The closest case is beta minus the OECD capability gap, -0.1115 with interval $[-0.2257, 0.0027]$; its paired MDE_{80} is 0.1689. Beta minus Webb AI is -0.0646 with interval $[-0.1682, 0.0389]$ and MDE_{80} 0.1499.

The correct conclusion is not that all architectures yield the same economic relationship, nor that a smaller alternative estimate shows the pattern “does not extend.” The design does not detect these paired differences and is too imprecise to rule out economically important movements in either direction. This is itself a result: the employment application reveals much

TABLE 4. Paired architecture differences are mostly unresolved

Alternative to Eloundou beta	Common occ.	$\hat{\beta}_{\text{beta}} - \hat{\beta}_m$	Paired SE	95% paired interval	MDE ₈₀
AIOE administrative	466	-.0717	.0454	[-.1614, .0181]	.1272
AIOE ability/direct	455	-.0273	.0446	[-.1134, .0589]	.1250
AIOE OEWS-weighted	456	-.0665	.0455	[-.1557, .0227]	.1273
Eloundou alpha	468	-.0324	.0376	[-.1054, .0406]	.1053
Eloundou broad	468	.0259	.0217	[-.0156, .0674]	.0609
Webb AI patent-task	448	-.0646	.0535	[-.1682, .0389]	.1499
OECD capability gap	448	-.1115	.0603	[-.2257, .0027]	.1689

Notes: Each row compares architecture-specific Q5-Q1 coefficients on the row's common support, using the same 9,999 occupation multipliers for both estimates. MDE₈₀ is the normal-theory minimum detectable paired difference at 80-percent power and a two-sided 5-percent level. Every interval includes zero. The data therefore do not detect these differences and do not establish economic equivalence or invariance. The six initial standardized scores are highly dependent: two weighted principal components explain 96.11 percent of their variance on the 444-occupation intersection.

more about how treatments are constructed than about which construction has a different population coefficient.

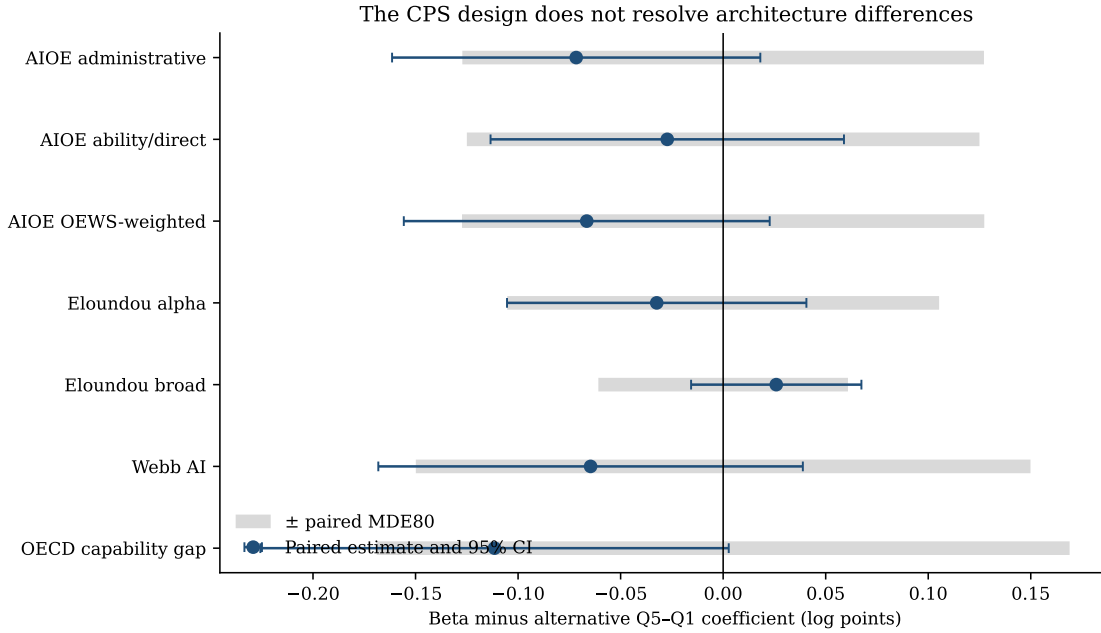


FIGURE 4. Paired differences and the resolution of the CPS design

Notes: Points are beta-minus-alternative Q5-Q1 coefficient differences on pair-specific common support. Thin lines are 95-percent common-draw paired intervals. Gray bars span plus or minus the normal-theory MDE at 80-percent power. An interval containing zero is a failure to detect a difference, not evidence of equivalence.

The original six-score intersection-union calculation is correspondingly demoted. All six marginal point estimates are negative on their literal intersection, and the maximum one-sided marginal tail area is .045, while the predeclared simultaneous-band criterion fails. With two principal components explaining 96.11 percent of score variance, this is a dependent-set sign statement, not six independent validations. Nothing in it overturns the paired-precision finding.

6.2 A direct bridge to the motivating result

The published component of [Brynjolfsson, Chandar, and Chen \(2026\)](#) that can be reproduced is a different representation: GPT-4 beta exposure divided into employment-weighted quintiles, with Q4-Q5 compared against Q1-Q3. Implementing that classification in the CPS model gives -0.0728 with interval $[-0.1240, -0.0216]$, roughly half the corrected Q5-Q1 log-point magnitude. The unconditioned result is -0.0728 as well. Young CPS stock rises 5.78 percent in the bottom-three group and falls 5.03 percent in the top-two group between November

2022 and June 2026, while older stocks rise by about 0.6 percent in each. These aggregate growth calculations do not inherit a hiring interpretation.

Reapplying the BCC grouping rule to eight exposure constructions produces native-support coefficients from -0.0817 for Eloundou broad to -0.0096 for the OECD capability gap. On literal 426-occupation common support, beta is -0.0810 and OECD is -0.0076 ; their paired difference is -0.0733 with interval $[-0.1390, -0.0076]$. The other six paired differences are not detected. This does not contradict Table 4: grouping changes the estimand and its precision. It illustrates that the economic contrast is part of the constructed empirical object.

TABLE 5. Changing the economic comparison

Comparison	Coef.	SE	95% interval	MDE ₈₀
Frozen beta Q5–Q1	-.1311	.0444	[-.2171, -.0451]	.1244
Beta Q5–Q2	-.0456	.0406	[-.1248, .0335]	.1137
Beta Q5–Q3	-.0833	.0387	[-.1581, -.0084]	.1081
Beta Q5–Q4	-.0340	.0393	[-.1110, .0429]	.1099
BCC beta top-two versus bottom-three	-.0728	.0260	[-.1240, -.0216]	.0728
Corrected beta Q5–Q1, quarterly cells	-.1345	.0450	[-.2222, -.0468]	.1260
Corrected beta Q5–Q1, respondent equivalents	-.1348	.0441	[-.2216, -.0481]	.1237

Notes: The first four rows are different contrasts from the same fitted quintile model, not changes of omitted-category notation. The BCC row reproduces only the public GPT-4 beta/top-two-versus-bottom-three component inside the CPS design; it does not reproduce proprietary ADP outcomes, job-title mapping, hiring margins, or firm-time controls. Quarterly aggregation and respondent-equivalent cells are post-outcome exploratory diagnostics. A confidence interval containing zero means the design does not detect a difference; it does not establish equivalence.

6.3 Taxonomy repair and construction layers

Taxonomy repair is the clearest measurement finding because it separates an invalid workflow from defensible alternatives. Literal matching of SOC-2010 AIOE codes into a post-2018 occupation universe covers only 3.33 percent of computer-and-mathematical employment. The documented route reaches 97.7 percent. In a continuous comparison, the original score on original support gives -0.01885 ; the repaired score on the same 410 occupations gives -0.01920 ; expanding to 495 repaired-score occupations gives -0.03156 ; excluding computer/math from the expanded support gives -0.02940 . Most movement comes from changing who enters, not correcting scores among already matched occupations.

This sequence is not a multiverse of equally defensible estimates. The exact-code merge is an implementation failure. The fixed-support row measures score correction. The expansion

changes the target population. The computer/math exclusion asks whether one family drives the expanded-support coefficient. Reporting these layers separately is the portable contribution.

6.4 Primitive models rather than rotated labels

The prior draft emphasized “consensus” and “disagreement” coordinates $F = (A + E)/2$ and $G = (A - E)/2$. Those coordinates are retained in the online appendix only as a complete record. A representative two-score model is more transparent: administrative AIOE enters at +0.0449 and beta at −0.0788 when included jointly. In direct task units, D and S enter at −0.0323 and −0.0303. These are conditional associations among correlated constructed regressors, not causal technology channels. Transforming F/G back to A/E reproduces the same fitted predictor exactly and supplies no additional evidence.

7 Timing, Weights, and Remaining Interpretations

7.1 Pre-AI pseudo-breaks and post-period dynamics

The authenticated extract begins in January 2017, so the literal request for monthly pseudo-breaks beginning in 2015 is infeasible. I estimate every feasible break in the fully pre-AI January 2017–December 2019 window and emphasize 12 dates with at least 12 retained months on each side. Under both pre-AI classifications and the frozen classification, the balanced pseudo-break distributions are centered near zero; none approaches −0.1311. Because the 12 estimates overlap and were designed after the headline was known, the plus-one empirical tail of 1/13 is a resolution floor, not a randomization p -value. Unbalanced early-window extremes are artifacts of having as little as one pre-break month.

The actual post path remains descriptive rather than a causal event study. A model with separate periods gives −0.1356 in 2023, −0.0783 in 2024, and −0.1629 in 2025–26. The last is distinguishable from 2024 but not from 2023. No detected pretrend and an end-2022 break do not isolate AI from the technology-sector correction, interest rates, return-to-office changes, post-pandemic normalization, or occupational composition.

I do not add a national adoption series merely to replace the post indicator. Interacting one national series with fixed occupation exposure would largely reparameterize an already displayed exposure-by-time path. Industry-level BTOS adoption would require an occupation–industry ecological bridge and would still not generate exogenous occupational adoption. Such data could support a separately designed adoption paper, but forcing it into this measurement-centered revision would change both treatment and estimand without resolving the main

confound.

7.2 Population controls and response composition

January 2025 introduced unusually large CPS population-control revisions. A literal age-by-occupation counterfactual series “without” those controls cannot be constructed: BLS provides aggregate experimental adjustments, not subgroup weights, and official earlier microdata are not revised. I therefore report what the available data identify rather than rescale cells ad hoc.

On the repaired calendar, official weights give -0.1346 and respondent-equivalent cells give -0.1348 . Ending in December 2024 gives -0.1135 and -0.1056 , respectively. In a joint official-weight model, 2023–24 is -0.1108 and 2025–26 is -0.1664 ; their paired difference is -0.0556 with interval $[-0.1228, 0.0116]$. Under respondent equivalents the era difference is more negative and detected. The raw official-weight Q5–Q1 log-ratio contrast becomes slightly less negative from December 2024 to January 2025. These checks do not provide a counterfactual weight vintage, but they are inconsistent with revised population weights mechanically creating the full association.

“Respondent equivalent” means that cells are reconstructed from observed employed records after route expansion without their CPS final weights. It changes the empirical cell outcome; dividing all weighted stocks by a common constant would not change the maximizer and is not what is done here.

7.3 Age, entry, and mobility

The pooled older comparison is not the sole source of the pattern. Ages 22–25 versus 26–35 give -0.1307 , nearly identical to the frozen pooled result. Comparisons with ages 26–45 and 36–55 are -0.1475 and -0.1464 . The contrast against ages 51–65 is smaller, -0.0978 , and its paired difference from the pooled comparison is detected. Age 22–25 is an age group, not observed experience; the paper therefore uses “young” rather than treating every respondent as a labor-market entrant.

The main outcome cannot determine whether hiring, separation, or occupation switching generated the stock association. CPS longitudinal links do allow distinct flow estimands, but their intervals are wide. Entry destination is an allocation conditional on moving from nonemployment to employment, not an employment-finding probability. The employed-to-employed switch frame says still less about hiring. For that reason, the mobility analysis, family-balanced disagreement statistics, rematching benchmark, and flow regressions appear in the online appendix rather than as a main contribution. No immediate-reversal rate is

treated as a validated occupation-coding-error rate.

8 Conclusion

Public CPS data reproduce a negative young-relative employment-stock association under a widely used occupational AI-exposure construction. After repairing the calendar, the top-versus-bottom beta-quintile contrast is -0.1346 log points. Implementing the public top-two-versus-bottom-three grouping used by [Brynjolfsson, Chandar, and Chen \(2026\)](#) gives a smaller but negative -0.0728 contrast. These are descriptive population comparisons, not replications of proprietary hiring results or causal AI estimates.

The decisive fact is where the comparison lies. Broad two-digit occupation-family controls reduce the detailed Q5–Q1 coefficient to about -0.032 . The paired movement is detectable, while the residual coefficient is imprecise and draws on only 30 percent of baseline target information. Only four broad families span both exposure tails. Thus, the baseline pattern is principally a comparison across broad occupational families; the design does not precisely isolate a detailed within-family AI gradient.

That conclusion should not be embellished into a mechanism the data reject. Low-exposure food-service occupations matter, but removing them attenuates the point estimate by less than one tenth. Broader in-person-service exclusions do not attenuate it. Joint influence exercises show large, oppositely signed occupation contributions rather than one occupation whose removal makes the result vanish. The evidence supports a broad-composition interpretation, not a demonstrated food-service-recovery explanation.

Nor do alternative exposure estimates permit a horse-race winner. All seven paired beta-minus-alternative Q5–Q1 intervals include zero, with minimum detectable differences as large as 0.169 log points. A confidence interval containing zero means that the design does not detect a difference; it does not establish economic equivalence. The original six implementations are also highly dependent, with two components accounting for 96.11 percent of their score variance. Claims of six independent validations or universal architecture robustness are therefore retired.

The measurement work remains useful precisely because it distinguishes objects that should not be pooled. Exact-code matching across incompatible occupational vintages is an implementation error. Repairing scores on fixed support is a measurement comparison. Readmitting occupations changes the population. Switching from Q5–Q1 to top-two-versus-bottom-three changes the estimand. Moving from LLM task acceleration to patent overlap changes the construct. Rotating F/G into A/E changes only coordinates. The revised tables report these layers rather than treating every row as interchangeable robustness evidence.

The paper’s claim is consequently narrower and stronger than before. The early-career CPS pattern is real as a descriptive broad-occupation contrast, but this design cannot attribute it specifically to detailed AI exposure. Future causal work needs observed adoption or deployment, a credible source of timing or assignment variation, and outcomes that directly measure hiring and separation. Future measurement work needs occupation-task validation labels collected independently of the constructed indices, repeated ratings or audited use logs, and a taxonomy-stable holdout sample. Until then, occupational AI coefficients should travel with their construct, mapping, support, comparison, and resolution limits.

References

- Acemoglu, Daron and Pascual Restrepo. 2019. “Automation and New Tasks: How Technology Displaces and Reinstates Labor.” *Journal of Economic Perspectives* 33 (2): 3–30. [10.1257/jep.33.2.3](#).
- Andrews, Isaiah, Matthew Gentzkow, and Jesse M. Shapiro. 2017. “Measuring the Sensitivity of Parameter Estimates to Estimation Moments.” *Quarterly Journal of Economics* 132 (4): 1553–1592. [10.1093/qje/qjx023](#).
- Autor, David H. and David Dorn. 2013. “The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market.” *American Economic Review* 103 (5): 1553–1597. [10.1257/aer.103.5.1553](#).
- Autor, David H., Frank Levy, and Richard J. Murnane. 2003. “The Skill Content of Recent Technological Change: An Empirical Exploration.” *Quarterly Journal of Economics* 118 (4): 1279–1333. [10.1162/003355303322552801](#).
- Battaglia, Laura, Timothy Christensen, Stephen Hansen, and Szymon Sacher. 2024. “Inference for Regression with Variables Generated by AI or Machine Learning.” CEPR Discussion Paper 19115, CEPR Press. <https://cepr.org/publications/dp19115>. Revised May 2025.
- Brynjolfsson, Erik, Bharat Chandar, and Ruyu Chen. 2026. “Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence.” , Stanford Digital Economy Lab. <https://digitaleconomy.stanford.edu/publication/canaries-in-the-coal-mine-six-facts-about-the-recent-employment-effects-of-artificial-intelligence/>. Revised August 12, 2026; original version published 2025.
- Brynjolfsson, Erik, Danielle Li, and Lindsey R. Raymond. 2025. “Generative AI at Work.” *Quarterly Journal of Economics* 140 (2): 889–942. [10.1093/qje/qjae044](#).
- Christensen, Timothy and Stephen Hansen. 2026. “Performing Valid Inference with AI/ML-Generated Covariates: A Guide for Empirical Practice.” *AEA Papers and Proceedings* 116: 92–97. [10.1257/pandp.20261020](#).
- Duan, Junting and Markus Pelger. 2026. “Inference with AI-Generated Covariates.” NBER Working Paper 35481, National Bureau of Economic Research. [10.3386/w35481](#).
- Eloundou, Tyna, Sam Manning, Pamela Mishkin, and Daniel Rock. 2024. “GPTs Are GPTs: Labor Market Impact Potential of LLMs.” *Science* 384 (6702): 1306–1308. [10.1126/science.adj0998](#).
- Felten, Edward W., Manav Raj, and Robert Seamans. 2018. “A Method to Link Advances in Artificial Intelligence to Occupational Abilities.” *AEA Papers and Proceedings* 108: 54–57. [10.1257/pandp.20181021](#).
- Felten, Edward W., Manav Raj, and Robert Seamans. 2021. “Occupational, Industry, and Geographic Exposure to Artificial Intelligence: A Novel Dataset and Its Potential Uses.” *Strategic Management Journal* 42 (12): 2195–2217. [10.1002/smj.3286](#).
- Humlum, Anders and Emilie Vestergaard. 2025. “Still Waters, Rapid Currents: Early Labor Market Transformation under Generative AI.” NBER Working Paper 33777, National Bureau of Economic Research. [10.3386/w33777](#). Revised March 2026.

- Ludwig, Jens, Sendhil Mullainathan, and Ashesh Rambachan. 2026. “Large Language Models: An Applied Econometric Framework.” *Annual Review of Economics* 18: 283–316. [10.1146/annurev-economics-120925-105620](#).
- Organisation for Economic Co-operation and Development. 2026. “The OECD AI Exposure Measure: Mapping the OECD AI Capability Indicators to Occupations.” OECD Artificial Intelligence Papers 59, OECD Publishing. [10.1787/f3da0f0a-en](#).
- Rambachan, Ashesh and Jonathan Roth. 2023. “A More Credible Approach to Parallel Trends.” *Review of Economic Studies* 90 (5): 2555–2591. [10.1093/restud/rdad018](#).
- Webb, Michael. 2020. “The Impact of Artificial Intelligence on the Labor Market.” , Stanford University. https://www.michaelwebb.co/webb_ai.pdf.